$$
\begin{aligned}
&C_{ud}^{(1)}\mathtt{i1\_i2\_i3\_i4\_} \to \\
&\frac{1}{16\pi^2}\Big(-\frac{8}{243}\sum_p g_1^4\,\mathrm{LF}_{3,0}\big[m_{\tilde d}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{81}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde d}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{32}{1215}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde d}{}^{p}\big] \\
&\quad \delta_{i1i2}\,\delta_{i3i4}-\frac{8}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}\big[m_{\tilde e}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{27}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde e}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{32}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde e}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}\big[m_{\tilde l}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{5}{54}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde l}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{16}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde l}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{4}{243}\sum_p g_1^4\,\mathrm{LF}_{3,0}\big[m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{162}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{16}{1215}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{32}{243}\sum_p g_1^4\,\mathrm{LF}_{3,0}\big[m_{\tilde u}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{20}{81}\sum_p g_1^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde u}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{128}{1215}\sum_p g_1^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde u}{}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{4}{81}\,g_1^4\,\mathrm{LF}_{3,0}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{27}\,g_1^4\,\mathrm{LF}_{4,-1}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{32}{405}\,g_1^4\,\mathrm{LF}_{5,-2}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{1}{243}\,g_1^4\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde d}{}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{243}\,g_1^4\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde d}{}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{2}{243}\,g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde d}{}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{243}\,g_1^4\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde d}{}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{1}{243}\,g_1^4\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde d}{}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{243}\,g_1^4\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde d}{}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{2}{243}\,g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde d}{}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{243}\,g_1^4\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde d}{}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{4}{243}\,g_1^4\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{243}\,g_1^4\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{243}\,g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{243}\,g_1^4\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{4}{243}\,g_1^4\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{243}\,g_1^4\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{243}\,g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{243}\,g_1^4\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_3,m_{\tilde d}{}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\big[m_3,m_{\tilde d}{}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_3,m_{\tilde d}{}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\big[m_3,m_{\tilde d}{}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_3,m_{\tilde d}{}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\big[m_3,m_{\tilde d}{}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_3,m_{\tilde d}{}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\big[m_3,m_{\tilde d}{}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_3,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\big[m_3,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_3,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\big[m_3,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad \frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_3,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\big[m_3,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_3,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\big[m_3,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{2}{243}\,g_1^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde d}{}^{i3},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{243}\,g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde d}{}^{i3},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde d}{}^{i3},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde d}{}^{i3},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{2}{243}\,g_1^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde d}{}^{i4},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{243}\,g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde d}{}^{i4},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde d}{}^{i4},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde d}{}^{i4},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{1}{27}\,g_1^2\left(-2\,\overline{y_d}^{pi3}\,y_d{}^{pi4}\,\delta_{i1i2}+\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\delta_{i3i4}\right)\mathrm{LF}_{2,1,0}\big[m_{\tilde q}{}^{p},\tilde\mu\big]+ \\
&\quad \frac{1}{54}\,g_1^2\left(2\,\overline{y_d}^{pi3}\,y_d{}^{pi4}\,\delta_{i1i2}-\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\delta_{i3i4}\right)\mathrm{LF}_{2,2,-1}\big[m_{\tilde q}{}^{p},\tilde\mu\big]+ \\
&\quad \frac{1}{54}\,g_1^2\left(2\,\overline{y_d}^{pi3}\,y_d{}^{pi4}\,\delta_{i1i2}-\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\delta_{i3i4}\right)\mathrm{LF}_{3,1,-1}\big[m_{\tilde q}{}^{p},\tilde\mu\big]+ \\
&\quad \frac{8}{243}\,g_1^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde u}{}^{i1},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{243}\,g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde u}{}^{i1},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde u}{}^{i1},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde u}{}^{i1},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{243}\,g_1^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde u}{}^{i2},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{243}\,g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde u}{}^{i2},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{8}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde u}{}^{i2},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde u}{}^{i2},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{1}{54}\,g_1^2\left(2\,\overline{y_d}^{pi3}\,y_d{}^{pi4}\,\delta_{i1i2}-\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\delta_{i3i4}\right)\mathrm{LF}_{2,1,0}\big[\tilde\mu,m_{\tilde q}{}^{p}\big]+ \\
&\quad \frac{1}{54}\,g_1^2\left(14\,\overline{y_d}^{pi3}\,y_d{}^{pi4}\,\delta_{i1i2}+11\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\delta_{i3i4}\right)\mathrm{LF}_{3,1,-1}\big[\tilde\mu,m_{\tilde q}{}^{p}\big]- \\
&\quad \frac{2}{27}\,g_1^2\left(\overline{y_d}^{pi3}\,y_d{}^{pi4}\,\delta_{i1i2}+\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\delta_{i3i4}\right)\mathrm{LF}_{4,1,-2}\big[\tilde\mu,m_{\tilde q}{}^{p}\big]+ \\
&\quad \frac{2}{81}\,g_1^4\,\mathrm{LF}_{2,1,1,-1}\big[m_1,m_{\tilde d}{}^{i3},m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{4}{81}\,g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}\big[m_1,m_{\tilde d}{}^{i3},m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{2}{81}\,g_1^4\,\mathrm{LF}_{2,1,1,-1}\big[m_1,m_{\tilde d}{}^{i4},m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{4}{81}\,g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}\big[m_1,m_{\tilde d}{}^{i4},m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{1}{9}\,g_3^4\,\mathrm{LF}_{2,1,1,-1}\big[m_3,m_{\tilde d}{}^{i3},m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{9}\,g_3^4\,m_3^2\,\mathrm{LF}_{2,1,1,0}\big[m_3,m_{\tilde d}{}^{i3},m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{1}{9}\,g_3^4\,\mathrm{LF}_{2,1,1,-1}\big[m_3,m_{\tilde d}{}^{i4},m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{9}\,g_3^4\,m_3^2\,\mathrm{LF}_{2,1,1,0}\big[m_3,m_{\tilde d}{}^{i4},m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad \frac{1}{6}\,\overline{y_d}^{pi3}\,y_d{}^{ri4}\,\overline{y_u}^{ri1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,-1}\big[\tilde\mu,m_{\tilde q}{}^{p},m_{\tilde q}{}^{r}\big]- \\
&\quad \frac{1}{3}\,\tilde\mu^2\,\overline{y_d}^{pi3}\,\overline{y_u}^{ri1}\left(y_d{}^{ri4}\,y_u{}^{pi2}-3\,y_d{}^{pi4}\,y_u{}^{ri2}\right)\mathrm{LF}_{2,1,1,0}\big[\tilde\mu,m_{\tilde q}{}^{p},m_{\tilde q}{}^{r}\big]\Big)
\end{aligned}
$$